

# Rainy Music Player

## Feature Testing Document

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Environment: <http://192.168.1.2:6969>

**9 New Features | 22 Automated Tests | All Passing**

## . Overview

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This document describes 9 new features added to the Rainy self-hosted music player, along with step-by-step manual testing instructions and expected results for each. All backend functionality is also covered by an automated pytest suite (tests/test\_new\_features.py) with 22 tests, all passing.

### Feature List

- [ ] 1. Playback History & Recently Played
- [ ] 2. PWA + Media Session (installable app, lock-screen controls)
- [ ] 3. Queue Management (add to queue, play next, clear queue)
- [ ] 4. Smart Mix / Auto-Mix / Radio Mode
- [ ] 5. Cross-Device Playback Sync
- [ ] 6. Keyboard Shortcuts + Global Search (Ctrl+K)
- [ ] 7. Drag-and-Drop Playlist Reordering
- [ ] 8. Album View (grid + detail page)
- [ ] 9. Sleep Timer with Volume Fade-Out

### How to Run Automated Tests

```
cd /opt/data/Rainy && .venv/bin/python -m pytest tests/test_new_features.py -v
```

Result: 22 passed in ~3s

## 1. Playback History & Recently Played

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Every song play is recorded to the play\_history table. A new 'Recently Played' nav item shows recently played songs, most-played songs, and listening statistics (total plays, unique songs, active days, listening time, top artists, top genres).

### Manual Test Steps

1. Log in and play 3-4 different songs from your library.

*Expected: Each play is recorded server-side (no visible change yet).*

2. Click 'Recently Played' in the left sidebar.

*Expected: The recently played list shows the songs you just played, most recent first, with timestamps like '2m ago'.*

3. Click the 'Most Played' tab.

*Expected: Songs are ranked by play count. The song you played most appears at the top with its play count.*

4. Click the 'Listening Stats' tab.

*Expected: Stat cards show Total Plays, Unique Songs, Active Days, and Listening Time. Top Artists and Top Genres lists are populated.*

5. Click 'Clear History' and confirm.

*Expected: A success toast appears and the recently played list becomes empty.*

## 2. PWA + Media Session

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Rainy is now an installable Progressive Web App with a manifest, service worker (offline shell caching), icons, and responsive mobile layout. The Media Session API exposes lock-screen / OS media controls (play, pause, next, previous, seek).

### Manual Test Steps

1. Open <http://192.168.1.2:6969> in Chrome/Edge on desktop.  
*Expected: An 'Install' icon/prompt appears in the address bar (app is installable).*
2. Install the app and launch it.  
*Expected: It opens in a standalone window (no browser chrome) with the Rainy icon.*
3. Play a song, then check your OS media controls / lock screen.  
*Expected: The song title, artist, and artwork appear. Play/pause/next/prev buttons control Rainy.*
4. Open the app on a phone browser and rotate / resize to a narrow width.  
*Expected: The layout adapts: sidebar collapses, grids reflow to fewer columns, player bar wraps.*
5. Open DevTools > Application > Service Workers.  
*Expected: The 'rainy-v1' service worker is registered and active.*

### 3. Queue Management

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The queue context menu now includes 'Play Next' and 'Add to Queue'. A dedicated queue button in the player bar opens the fullscreen player's queue tab. `clearQueue` is available programmatically and via the queue UI.

#### Manual Test Steps

1. Play a song, then open the fullscreen player (click the artwork or press F).  
*Expected: The fullscreen player opens. Switch to the Queue tab.*
2. Right-click (or use the menu) on a queued song.  
*Expected: The context menu shows Play, Play Next, Add to Queue, Like, Dislike, Remove.*
3. Choose 'Play Next' on a song.  
*Expected: That song moves to the position immediately after the current song.*
4. Click the queue icon in the bottom player bar.  
*Expected: The fullscreen player opens directly to the Queue tab.*

## 4. Smart Mix / Auto-Mix / Radio Mode

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A new 'Smart Mix' nav item offers three auto-generated queue modes: Radio (songs similar to the current track by genre/artist affinity), Liked Mix (shuffle of liked songs), and Discovery (tracks you haven't played yet). Results can be played individually or all at once.

### Manual Test Steps

1. Click 'Smart Mix' in the sidebar.

*Expected: Three mode cards appear: Radio, Liked Mix, Discovery.*

2. Click 'Discovery'.

*Expected: A list of up to 30 unplayed songs appears with a 'Play All' button.*

3. Click 'Play All'.

*Expected: Playback starts from the first song in the generated mix; the queue is populated with the rest.*

4. Play a song, return to Smart Mix, and click 'Radio'.

*Expected: A mix of songs sharing genre/artist affinity with the playing song is generated.*

5. Click an individual song row in the mix.

*Expected: That song starts playing with the rest of the mix as the queue.*

## 5. Cross-Device Playback Sync

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Playback state (current song, position, queue, is\_playing) is saved to the server (playback\_state table) and can be restored on another device, so you can pause on one device and resume on another.

### Manual Test Steps

1. On device A, play a song and seek to ~1:00, then pause.

*Expected: State is saved to the server (song, position, queue).*

2. On device B (another browser/phone), log in and load Rainy.

*Expected: The app offers to restore the last playback state (or auto-restores the queue/position).*

3. Resume playback on device B.

*Expected: The same song resumes near the saved position with the same queue.*

## 6. Keyboard Shortcuts + Global Search (Ctrl+K)

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Global keyboard shortcuts: Space = play/pause, Left/Right = seek 5s, Up/Down = volume, N/P = next/prev, M = mute, F = fullscreen, L = like, / = focus search. Ctrl+K (or Cmd+K) opens a command-palette-style global search overlay that searches songs, artists, albums, and playlists simultaneously.

### Manual Test Steps

1. Press Space while a song is loaded.  
*Expected: Playback toggles play/pause.*
2. Press the Right arrow.  
*Expected: The track seeks forward ~5 seconds.*
3. Press Ctrl+K (Cmd+K on Mac).  
*Expected: A centered search overlay appears with a text input and grouped results.*
4. Type a song, artist, or album name.  
*Expected: Matching results appear grouped by type (Songs, Artists, Albums, Playlists).*
5. Click a song result.  
*Expected: The overlay closes and that song starts playing.*
6. Press Escape.  
*Expected: The overlay closes.*

## 7. Drag-and-Drop Playlist Reordering

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In a playlist's list view, song rows are draggable. Drag a song to a new position and release; the new order is persisted to the backend (order\_num) and survives reloads.

### Manual Test Steps

1. Open a playlist with 3+ songs and switch to List view.

*Expected: Song rows show a grab cursor on hover.*

2. Drag the 3rd song up to the 1st position and release.

*Expected: The row moves live during drag; on release a 'Playlist order saved' toast appears.*

3. Reload the page and reopen the playlist.

*Expected: The song order is preserved (the moved song is still first).*

## 8. Album View

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A new 'Albums' nav item shows a searchable, sortable grid of all albums (by name, artist, year, or recently added). Clicking an album opens a detail page with cover art, metadata, a 'Play Album' button, and the full tracklist.

### Manual Test Steps

1. Click 'Albums' in the sidebar.

*Expected: A grid of album cards appears, each with cover art, title, artist, and song count.*

2. Type in the album search box.

*Expected: The grid filters to matching albums by name or artist.*

3. Change the sort dropdown to 'Artist'.

*Expected: Albums re-sort alphabetically by artist.*

4. Click an album card.

*Expected: The album detail page opens with large cover art, title, artist, year, total duration, a 'Play Album' button, and the numbered tracklist.*

5. Click 'Play Album' or any track row.

*Expected: Playback starts with the full album as the queue.*

6. Click 'Back to Albums'.

*Expected: You return to the album grid.*

## 9. Sleep Timer with Volume Fade-Out

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A clock icon in the player bar opens a sleep timer menu with presets (5/10/15/30/45/60 min) and a custom input. When the timer ends, volume fades out over ~10 seconds and playback pauses. The button shows a live countdown while active.

### Manual Test Steps

1. Play a song, then click the clock (sleep timer) icon in the player bar.  
*Expected: A popup menu appears with preset durations and a custom input.*
2. Click '5 min' (or set a short custom value like 1 min for testing).  
*Expected: A success toast confirms the timer; the clock icon highlights and shows the countdown.*
3. Wait for the timer to elapse.  
*Expected: Volume gradually fades out, then playback pauses. A toast says 'Sleep timer ended'. The clock icon returns to normal.*
4. Start a timer again, then reopen the menu and click 'Cancel'.  
*Expected: The timer is cancelled, the countdown stops, and playback continues unaffected.*

## . Automated Test Summary

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File: tests/test\_new\_features.py | Framework: pytest | Result: 22 passed

### Playback History

test\_record\_and\_get\_history, test\_history\_requires\_song\_id, test\_history\_stats, test\_top\_artists, test\_clear\_history, test\_history\_requires\_auth

### Cross-Device State

test\_save\_and\_get\_state, test\_state\_overwrites, test\_state\_requires\_auth

### Albums

test\_list\_albums, test\_album\_search, test\_album\_detail, test\_albums\_requires\_auth

### Smart Mix

test\_smartmix\_discovery, test\_smartmix\_radio, test\_smartmix\_liked, test\_smartmix\_invalid\_mode, test\_smartmix\_requires\_auth

### Playlist Reorder

test\_playlist\_reorder, test\_playlist\_reorder\_empty\_list, test\_playlist\_reorder\_not\_found, test\_playlist\_reorder\_requires\_auth